Azith

(Powder for Oral Suspension 200mg/5mL)

PRODUCT NAME
Azith (Powder for Oral Suspension 200mg/5mL).

NAME AND STRENGTH OF ACTIVE SUBSTANCE AND EXCIPIENT
Each 5 mL (after reconstituted) contains Azithromycin dihydrate equivalent to Azithromycin 200 mg
Each bottle contains Azithromycin 600 mg/15 mL.

DOSAGE FORM
Dry powder for oral suspension.

PRODUCT DESCRIPTION
Off-white powder with characteristic odour (vanilla and orange).
After reconstitution, it yields a white to slightly yellow coloured suspension.

PHARMACODYNAMICS
<u>Mode of action:</u>
AZITH is a macrolide antibiotic belonging to the azalide group. The molecule is constructed by adding a nitrogen atom to the lactone ring of erythromycin A. The chemical name of azithromycin is 9-deoxy-9a-aza-9a-methyl-9a-homoerythromycin A. The molecular weight is 749.0. The mechanism of action of azithromycin is based upon the suppression of bacterial protein synthesis by means of binding to the ribosomal 50S sub-unit and inhibition of peptide translocation.
<u>Mechanism of resistance:</u>
Resistance to azithromycin may be inherent or acquired. There are three main mechanisms of resistance in bacteria: target site alteration, alteration in antibiotic transport and modification of the antibiotic.
Azithromycin demonstrates cross resistance with erythromycin resistant gram positive isolates. A decrease in macrolide susceptibility over time has been noted particularly in *Streptococcus pneumoniae* and *Staphylococcus aureus.* Similarly, decreased susceptibility has been observed among *Streptococcus viridans* and *Streptococcus agalactiae* (Group B) streptococcus against other macrolides and lincosamides.
Susceptibility
The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.
Table: Antibacterial spectrum of Azithromycin

Commonly susceptible species
Aerobic Gram-positive microorganisms
Staphylococcus aureus Methycillin-susceptible
Streptococcus pneumoniae Penicillin-susceptible
Streptococcus pyogenes **(Group A)**
Aerobic Gram-negative microorganisms
Haemophilus influenzae *Haemophilus parainfluenzae*
Legionella pneumophila
Moraxella catarrhalis
Neisseria gonorrhoeae
Pasteurella multocida
Anaerobic microorganisms
Clostridium perfringens
Fusobacterium spp.
Prevotella spp.
Porphyromonas spp.
Other microorganisms
Chlamydia trachomatis
Species for which acquired resistance may be a problem
Aerobic Gram-positive microorganisms
Streptococcus pneumoniae Penicillin-intermediate Penicillin-resistant
Inherently resistant organisms
Aerobic Gram-positive microorganisms
Enterococcus faecalis
Staphylococci **MRSA, MRSE***
Anaerobic microorganisms
Bacteroides fragilis group

* Methycillin-resistant staphylococci have a very high prevalence of acquired resistance to macrolides and have been placed here because they are rarely susceptible to azithromycin.

PHARMACOKINETICS
<u>Absorption</u>
Bioavailability after oral administration is approximately 37%. Peak plasma concentrations are attained 2 to 3 hours after taking the medicinal product.
<u>Distribution</u>
Orally administered azithromycin is widely distributed throughout the body.
In pharmacokinetic studies it has been demonstrated that the concentrations of azithromycin measured in tissues are noticeably higher (as much as 50 times) than those measured in plasma, which indicates that the agent strongly binds to tissues. Binding to serum proteins varies according to plasma concentration and ranges from 12% at 0.5 microgram/mL up to 52% at 0.05 microgram azithromycin/mL serum. The mean volume of distribution at steady state (Vss) has been calculated to be 31.1 L/kg.
<u>Elimination</u>
The terminal plasma elimination half-life closely reflects the elimination half-life from tissues of 2 to 4 days.
Approximately 12% of an intravenously administered dose of azithromycin is excreted unchanged in urine within the following three days. Particularly high concentrations of unchanged azithromycin have been found in human bile. Also in bile, ten metabolites were detected, which were formed through N- and O-demethylation, hydroxylation of desosamine and aglycone rings and cleavage of cladinose conjugate. Comparison of the results of liquid chromatography and microbiological analyses has shown that the metabolites of azithromycin are not microbiologically active.

INDICATIONS
Azithromycin is indicated for infections caused by susceptible organisms; in lower respiratory tract infections including bronchitis and pneumonia, in skin and soft tissue infections, in acute otitis media and in upper respiratory tract infections including sinusitis and pharyngitis/tonsillitis. (Penicillin is the usual drug of choice in the treatment of *Streptococcus pyogenes* pharyngitis, including the prophylaxis of rheumatic fever. Azithromycin is generally effective in the eradication of streptococci from the oropharynx, however, data establishing the efficacy of azithromycin and the subsequent prevention of rheumatic fever are not available at present.) In sexually transmitted diseases in men and women, azithromycin is indicated in the treatment of uncomplicated genital infections due to *Chlamydia trachomatis*. It is also indicated in the treatment of chancroid due to *Haemophilus ducreyi*; and uncomplicated genital infection due to non-multiresistant *Neisseria gonorrhoea*; concurrent infection with *Treponema pallidum* should be excluded.

RECOMMENDED DOSE
<u>Azithromycin Oral</u>
Oral azithromycin should be administered as a single daily dose. The period of dosing with regard to infection is given below.
Azithromycin powder for oral suspension can be taken with or without food.
<u>In adults</u>
For the treatment of sexually transmitted diseases caused by *Chlamydia trachomatis* and *Haemophilus ducreyi*, or susceptible *Neisseria gonorrhoeae*, the dose is 1000 mg as a single oral dose.
For patients who are allergic to penicillin and/or cephalosporins, prescribers should consult local treatment guidelines.
<u>In children</u>
The maximum recommended total dose for any treatment is 1500 mg for children. In general, the total dose in children is 30 mg/kg. Treatment for pediatric streptococcal pharyngitis should be dosed at a different regimen (see below). The total dose of 30 mg/kg should be given as a single daily dose of 10 mg/kg daily for 3 days, or given over 5 days with a single daily dose of 10 mg/kg on Day 1, then 5 mg/kg on Days 2-5.
As an alternative to the above dosing, treatment for children with acute otitis media can be given as a single dose of 30 mg/kg.

<u>For pediatric streptococcal pharyngitis</u>, azithromycin given as a single dose of 10 mg/kg or 20 mg/kg for 3 days has been shown to be effective; however, a daily dose of 500 mg must not be exceeded. In clinical trials comparing these two dosage regimens, similar clinical efficacy was observed but greater bacteriologic eradication was evident at the 20 mg/kg/day dose. However, penicillin is the usual drug of choice for the treatment of *Streptococcus pyogenes* pharyngitis, including prophylaxis of rheumatic fever.
For children weighing less than 15 kg, azithromycin suspension should be measured as closely as possible. For children weighing 15 kg or more, azithromycin suspension should be administered according to the guide provided below.

Azithromycin Suspension 30 mg/kg Total Treatment Dose		
Weight (kg)	**3-Day Regimen**	**5-Day Regimen**
< 15	10 mg/kg once daily on Days 1-3.	10 mg/kg on Day 1, then 5 mg/kg once daily on Days 2-5.
15-25	200 mg (5 mL) once daily on Days 1-3.	200 mg (5 mL) on Day 1, then 100 mg (2.5 mL) once daily on Days 2-5.
26-35	300 mg (7.5 mL) once daily on Days 1-3.	300 mg (7.5 mL) on Day 1, then 150 mg (3.75 mL) once daily on Days 2-5.
36-45	400 mg (10 mL) once daily on Days 1-3.	400 mg (10 mL) on Day 1, then 200 mg (5 mL) once daily on Days 2-5.
> 45	Dose as per adults.	Dose as per adults.

Special populations:
<u>In the Elderly</u>
The same dosage as in adult patients is used in the elderly. Elderly patients may be more susceptible to the development of torsades de pointes arrhythmia than younger patients.
<u>In Patients with Renal Impairment</u>
No dose adjustment is necessary in patients with mild to moderate renal impairment (GFR 10-80 mL/min). Caution should be exercised when azithromycin is administered to patients with severe renal impairment (GFR < 10 mL/min).
<u>In Patients with Hepatic Impairment</u>
The same dosage as in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

ROUTE OF ADMINISTRATION
Oral

CONTRAINDICATION
The use of this product is contraindicated in patients with a known hypersensitivity to azithromycin, erythromycin, any macrolide or ketolide antibiotic, or to any of the excipients

WARNINGS AND PRECAUTIONS
<u>Hypersensitivity</u>
In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)], **AZITH** should be discontinued immediately and appropriate treatment should be urgently initiated.
<u>Prolongation of the QT interval</u>
Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin. Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:
- Patients with congenital or documented QT prolongation
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones
- Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesemia
- Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency
- Elderly patients: elderly patients may be more susceptible to drug-associated effects on the QT interval

Actual Size 100 %

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Infantile hypertrophic pyloric stenosis (IHPS)
Infantile hypertrophic pyloric stenosis (IHPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and/ or irritability with feeding occurs.
Hepatotoxicity
Since the liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease. Cases of fulminant hepatitis potentially leading to life-threatening liver failure have been reported with azithromycin. Some patients may have had pre-existing hepatic disease or may have been taking other hepatotoxic medicinal products. In case of signs and symptoms of liver dysfunction, such as rapid developing asthenia associated with jaundice, dark urine, bleeding tendency or hepatic encephalopathy, liver function tests / investigations should be performed immediately.
Azithromycin administration should be stopped if liver dysfunction has emerged.
Ergot derivatives
In patients receiving ergot derivatives, ergotism has been precipitated by co-administration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be co-administrated.
Superinfection
As with any antibiotic preparation, observation for signs of superinfection with non-susceptible organisms including fungi is recommended.
Clostridium difficile associated diarrhoea
Clostridium difficile associated diarrhoea (CDAD) has been reported with the use of nearly all antibacterial agents, including azithromycin, and may range in severity from mild diarrhoea to fatal colitis. Strains of *C. difficile* producing hypertoxin A and B contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. Therefore, CDAD must be considered in patients who present with diarrhoea during or subsequent to the administration of any antibiotics. Careful medical history is necessary since CDAD has been reported to occur over 2 months after the administration of antibacterial agents. Discontinuation of therapy with azithromycin and the administration of specific treatment for *C. difficile* should be considered.
Streptococcal infections
Penicillin is usually the first choice for treatment of pharyngitis/tonsillitis due to *Streptococcus pyogenes* and also for prophylaxis of acute rheumatic fever. Azithromycin is in general effective against streptococcus in the oropharynx, but no data are available that demonstrate the efficacy of azithromycin in preventing acute rheumatic fever.
Renal impairment
In patients with severe renal impairment (GFR <10 mL/min) a 33% increase in systemic exposure to azithromycin was observed
Myasthenia gravis
Exacerbations of the symptoms of myasthenia gravis and new onset of myasthenia syndrome have been reported in patients receiving azithromycin therapy.
Diabetes
Caution in diabetic patients: 5 mL of reconstituted suspension contains 5.195 g of sucrose. Patients with rare hereditary problems of fructose intolerance, glucose-galactose malabsorption or sucrase-isomaltase insufficiency should not take this medicine.

INTERACTIONS WITH OTHER MEDICAMENTS

Antacids
In patients receiving both oral azithromycin and antacids, the drugs should not be taken simultaneously.
Digoxin
Concomitant administration of macrolide antibiotics, including azithromycin, with P-glycoprotein substrates such as digoxin, has been reported to result in increased serum levels of the P-glycoprotein substrate. Therefore, if azithromycin and P-glycoprotein substrates such as digoxin are administered concomitantly, the possibility of elevated serum digoxin concentrations should be considered. Clinical monitoring, and possibly serum digoxin levels, during treatment with azithromycin and after its discontinuation are necessary.
Zidovudine
Coadministration of zidovudine and azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells.
Ergot derivatives
Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.
Coumarin-type oral anticoagulants
There have been reports of potentiated anticoagulation subsequent to coadministration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.
Cyclosporin
Caution should be exercised before considering concurrent administration of cyclosporin and azithromycin. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.
Fluconazole
Coadministration of a single dose of 1200 mg azithromycin was not reported to alter the pharmacokinetics of a single dose of 800 mg fluconazole. However a clinically insignificant decrease in Cmax of azithromycin was reported.
Nelfinavir
Coadministration of azithromycin and nelfinavir reported to increase azithromycin concentrations. No significant adverse effects were reported and no dose adjustment is required.

PREGNANCY AND LACTATION

There are no adequate and well-controlled studies in pregnant women. Azithromycin should be used during pregnancy only if clearly needed.
There are no data on secretion in breast milk. As many drugs are excreted in human milk, azithromycin should not be used in the treatment of a lactating woman unless the physician feels that the potential benefits justify the potential risks to the infant.

SIDE EFFECTS

Azithromycin side effects are generally well tolerated. The adverse reactions listed below are classified by system organ and frequency.

System Organ	Frequency	Adverse Reactions
Infections and Infestations	Uncommon	Candidiasis, oral candidiasis, vaginal infection
	Not known	Pseudomembranous colitis
Blood and Lymphatic System Disorders	Uncommon	Leukopenia, neutropenia
	Not known	Thrombocytopenia, haemolytic anaemia
Immune System Disorders	Uncommon	Angioedema, hypersensitivity
	Not known	Anaphylactic reaction
Metabolism and Nutrition Disorders	Common	Anorexia
Nervous System Disorders	Common	Dizziness, headache, paraesthesia, dysgeusia
	Uncommon	Hypoaesethesia, somnolence, insomnia
	Not known	Myasthenia gravis
Eye Disorders	Common	Visual impairment
Ear and Labyrinth Disorders	Common	Deafness
	Uncommon	Hearing impaired, tinnitus
	Rare	Vertigo
Cardiac Disorders	Not known	Post-marketing experience Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongation and torsades de pointes (see *Warnings and Precautions*).
Gastrointestinal Disorders	Very common	Diarrhea, abdominal pain, nausea, flatulence
	Common	Vomiting, dyspepsia
	Uncommon	Gastritis, constipation
	Not known	Post-marketing experience Infantile hypertrophic pyloric stenosis
Hepatobiliary Disorders	Uncommon	Hepatitis
	Rare	Hepatic function abnormal
Skin and Subcutaneous Tissue Disorders	Not known	Post-marketing experience Severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)
Musculoskeletal, Connective Tissue Disorders	Common	Arthralgia
Renal and Urinary Disorders	Not known	Renal failure acute, nephritis interstitial
General disorders and Administration Site Conditions	Common	Fatigue
	Uncommon	Chest pain, oedema, malaise, asthenia

SYMPTOMS AND TREATMENT OF OVERDOSE

Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. The typical symptoms of an overdose with macrolide antibiotics include reversible loss of hearing, severe nausea, vomiting and diarrhea. In the event of overdose, the administration of medicinal charcoal and general symptomatic treatment and supportive measures are indicated as required.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

There is no evidence to suggest that azithromycin may have an effect on the patient's ability to drive or operate machinery.

INSTRUCTIONS FOR USE

At time of use, the dry powder should be reconstituted to form an oral suspension as below:
- Tap bottle to loosen powder
- Use the measuring cup provided, add 9 mL of water to the bottle and shake well
- Shake well before taking each dose

STORAGE CONDITIONS

The dry powder should be stored in unopened container in a dry place below 30°C. The reconstituted suspension should be stored below 30°C and used within 10 days. Do not freeze and discard any unused suspension after 10 days.

SHELF LIFE

Refer to unit carton box for the expiry date.
The reconstituted suspension should be used within 10 days.

DOSAGE FORMS AND PACKAGING AVAILABLE

AZITH powder for oral suspension 600 mg in amber glass 60 mL bottle (type III) with aluminium screw cap (line with PVC compound) in a carton box.
Pack contains clear transparent polypropylene measuring cup of 10 mL capacity having marking of 2.5 mL, 5 mL and 9 mL.

DATE OF REVISION OF PI

Date of revision: 17/02/2023

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Actual Size 100 %

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